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Diagnostics-VF.CONCEPT

SFRS for Zefiro-Europe

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Content

1 Introduction 3

1.1 Train Configuration 3

1.2 Definitions and abbreviations 3

2 System Overview 4

2.1 System Configuration 4

3 Functional requirements 5

3.1 Diagnostics 5
3.1.1 Function General requirements 5
3.1.2 Function Event handling 6
3.1.3 Function Event categorization 9
3.1.4 Buzzer interface 11
3.1.5 Function Condition data handling 11
3.1.6 Diagnostics data set at train level 12

3.2 Function Software version handling 13
3.2.1 Function Design 13
3.2.2 Requirements 13

3.3 Maintenance 14
3.3.1 Service Access 14
3.3.2 Software Download 15
3.3.3 Function Upload and analysis 15
3.3.4 Function Subscription of signals 16

4 Signal Interface 18

4.1 I/O Interface 18
4.1.1 Supplier system 18
4.1.2 TCMS SW 18

4.2 Bus Interface 18

5 References 19

6 Revision Log 20

6.1 Revision log 3EST000225-7918_B 20

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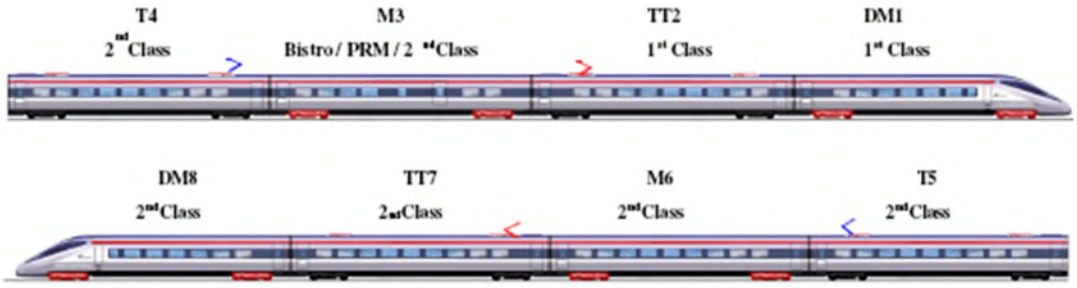
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1 Introduction

This document is a function specification for the control of the *Diagnostic* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

CD	Circuit Diagram
CCU	Central Computing Unit
CCU-D	Central Computing Unit - Diagnostic
CCU-O	Central Computing Unit - Operational
CSS	Common Software Structures (Operation system of CCU-O)
ECN	Ethernet Consist Network
EVENT	Diagnostics that will be stored in the Event database within TCMS system.
GUI	Graphical User Interface
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
IPT	Internet Protocol - Train
MAVIS	Maintenance of Vehicles using Information and Statistics
MTVD	MITRAC CC Tools Version control and Download
ODBS	Onboard Diagnostic Database System
PCL	Process Client
TCL	Tool Client
TCMS	Train Control and Management System
TDS	Train Diagnostic System
TSSP	Train Subscription Service Provider
VRS	Version Reporting System
WBS	Work Breakdown Structure

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2 System Overview

The purpose of the TCMS TDS (Train Diagnostic System) is to ascertain and provide the current status of the vehicle by gathering status data among the vehicle systems. Diagnostic messages are generated from both TCMS equipment as well as sub-suppliers equipment and stored by the diagnostic system. In case an event occurs which causes an operational impact to the vehicle it will be presented to the user combined with user advice.

2.1 System Configuration

The on-board part of the train diagnostic system, TDS, is a sub-system of the TCMS. The TCMS TDS located in CCU-D collects, stores and acts upon relevant information about events, faults and vehicle status from the connected systems. The event information stored in the TDS database can be presented by the TCMS IDU and can be uploaded to an off-board maintenance and service system for long term storage and additional off line analysis.

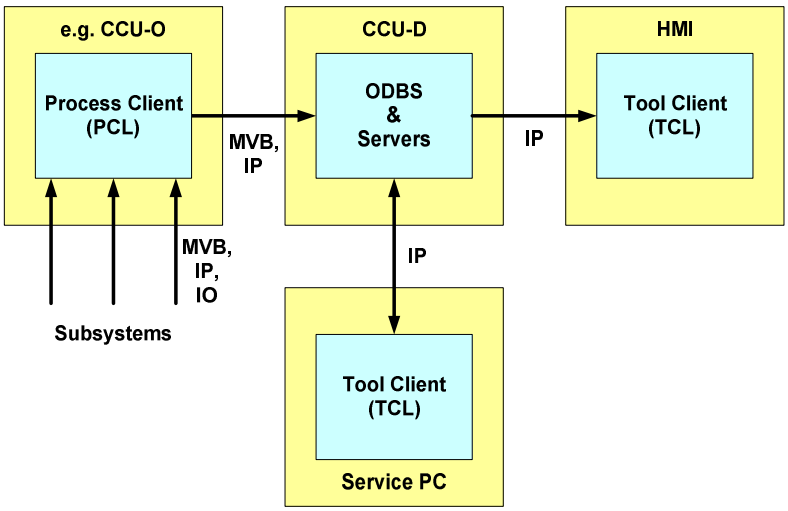


Figure: Train Diagnostic System (TDS)

For failure mode and redundancy concept refer to SFRS TCMS Operational.

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3 Functional requirements

3.1 Diagnostics

3.1.1 Function General requirements

3.1.1.1 Function Design

The TCMS diagnostic system stores the events defined in other SFRS documents. As events are detected both in the TCMS, as well as in the propulsion system, the scope split is defined.

3.1.1.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.66
Description	Events shall cover all systems listed in below including TCMS itself: – Air Supply System – ATP System – Auxiliary Electric System – - Battery System – - Brake System – - CCTV System – Driver's Safety Device – Exterior Doors – Fire Detection System – Heating, Ventilation and Air Conditioning System – High Voltage System – Infotainment – Interior Doors – Interior Lighting – Passenger Information System – Propulsion System – Toilets – Bogie Monitoring and other systems.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.120	The TCMS system shall implement the event reporting for the events defined in the different systems connected to TCMS and defined in the system SFRSes (except for events from the propulsion system).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.121	The propulsion system shall implement IP-PCL to report the events directly to the TCMS ODBS. The PCL and the ODBS shall use TDS2 including the redundancy concept.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.122	TDS files for the propulsion system events shall be delivered to the TCMS team for integration into the TDS configuration.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.123	The propulsion system TDS files shall be integrated into the TDS configuration.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.96
Description	Gathered event data and condition data shall be stored in a database located in battery backed up memory or with equivalent security so it can be used for operational, maintenance or vehicle performance measurements purposes.
Safety level	N/A

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ID	2F_10.01.Zefiro-Europe.TRS.129
Description	The diagnostic system shall be redundant.
Safety level	N/A

CCU-D / ODBS Firmware

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.146	The ODBS shall be configured to be stored in non volatile memory.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.322	Redundant ODBS shall be supported to maintain functionality under any single point of failure.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

ID	2F_10.01.Zefiro-Europe.TRS.100
Description	It shall be possible to clear the TDS Event database without uploading the database first.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.155	It shall be possible to clear the TDS ODBS without uploading the database first.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.1.2 Function Event handling

3.1.2.1 Function Design

Relating to the events a set of attributes needs to be defined that controls how events can be presented on the IDU as well as the descriptive text elements associated with the events.

3.1.2.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.103
Description	Each event shall have the following attributes: – Code/identity – Category – Start time – End time – Number of occurrences – - Texts
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.92
Description	The events shall be classified based on severity and character.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.98
Description	TDS shall support multiple language (namely French, German, Italian, English) event texts.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.99
Description	Italian shall be the default language. It shall be possible to switch between languages in run-time.
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.161	The instruction "MLN_3EVT12158-VZI TDS Process" shall be used.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

Event priority

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.163	For each event a priority shall be set (except for the propulsion system events). Note: TCMS team responsible for first proposal.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.164	For each propulsion system event a priority shall be set. Note: Propulsion team responsible for first proposal.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.165	Each event shall be classified based on severity, character and customer expectations.	Derived Requirement	5X_DOC.-.Documentation and Training		N/A

Event texts

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.140	TCMS TDS shall be configured for four different languages (French, German, Italian, English).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.150	For each event a short event description, extended event description (page 1) and a maintainer event description (page 2) shall be defined. Note: TCMS team responsible for first proposal of the short event description.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.152	For each propulsion event a short event description, extended event description (page 1) and a maintainer event description (page 2) shall be defined. Note: Propulsion team responsible for first proposal of the short event description.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.153	The short event description, extended event description (page 1) and a maintainer event description (page 2) shall be defined in French, German, Italian and English according to agreement with the customer.	Derived Requirement	5X_DOC.-.Documentation and Training		N/A

Event code / identity

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.293	Each TCMS event shall have a defined process ID and a defined ErrorCode0. TCMS shall not use process ID in the range 180 to 255 and shall not use ErrorCode0 in the range 40000 - 59999 that are used by propulsion system.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.294	Each propulsion event shall have a defined process ID in the range 180 to 255 and the ErrorCode0 in the range 40000 - 59999.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.295	Each TCMS event shall have an event code that is unique within each consist diagnostic database.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.296	Each propulsion event shall have an event code that is unique within each consist diagnostic database.	Derived Requirement	4S_05.-.Propulsion		N/A

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2F_10.01.Zefiro-Europe.CONCEPT.297	Subsystem groups shall be structured to fit the needs for the IDU active event overview menu.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

General event attributes

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.299	Events in the TDS ODBS shall be stored with start time, end time and number of occurrences.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

ID	2F_10.01.Zefiro-Europe.TRS.101
Description	The TDS module shall provide the global environmental signals that are generic for all events.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.68
Description	Environmental data with pre- and post-history shall be attatched to each group of events.
Safety level	N/A

Environment data

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.157	A set of environment data shall be defined and used for all generated events, with exception of propulsion system. Refer to “MLN_3EVT12158-VZI TDS Process”.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.169	A set of environment data shall be defined and used for all generated propulsion system events.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.170	Number of pre- and post samples of environmental data shall be configured per PCL.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.97
Description	The location and origin of collected data shall be visible. It shall be possible to determine car and unit.
Safety level	N/A

Event location

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.148	The PCL Location ID shall be set to values depending on car for the reported TCMS events. Car Location ID DM1 1 TT2 2 M3 3 T4 4 T5 5 M6 6 TT7 7 DM8 8	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_10.01.Zefiro-Europe.CONCEPT.302	The PCL Location ID shall be set to values depending on car for the reported propulsion events. Car Location ID DM1 1 TT2 2 M3 3 T4 4 T5 5 M6 6 TT7 7 DM8 8	Derived Requirement	4S_05.-.Propulsion		N/A

3.1.3 Function Event categorization

3.1.3.1 Function Design

Events are assigned to different categories. These categories are also used to control the behaviour of how events are presented on the IDU (as described in SFRS Visualization).

3.1.3.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.70
Description	Events shall use the following categories listed in priority order: 0. Serious falt events Information to the driver about the serious fault that has occurred. Automatically displayed on IDU when it has occured. 1. Advisory Information to the driver that he has done an incorrect maneuver. No acknowledge for this event. Automatically displayed on IDU when it has occured. 2. Alarm Major event, i.e. stopping events that require the relevant user's full attention. Automatically displayed on IDU when it has occured. 3. Warning Warnings or service effecting event, i.e. train performance reduction event. Automatically displayed on IDU when it has occured. 4. Maintenance Minor event intended for maintenance personnel. No service effecting event. Available in event lists on IDU for maintainers.. 5. Other Informative Event, Non effecting event. Available in event history list on IDU.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.105
Description	Events with higher priority shall take precedence over events with lower priority, even if a lower priority event has already been presented on the IDU.
Safety level	N/A

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_10.01.Zefiro-Europe.CONCEPT.224	Events shall be categorized into following event groups: Serious fault events, Advisory, Alarm, Warning, Maintenance, Other, with the priorities set according to (from highest priority to lowest): Serious fault events Advisory Alarm Warning Maintenance Other	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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2F_10.01.Zefiro-Europe.CONCEPT.223	For test purposes one event of each type shall be implemented in a way (connected to a software parameter) that makes it easy to set the events in a test set-up.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.131
Description	Serious fault events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU. When the event has been acknowledged it shall be removed from the screen.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.341	Serious fault events shall be presented on the IDU and shall be acknowledged using a button on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_10.01.Zefiro-Europe.TRS.106
Description	Advisory event shall be displayed on the IDU. As soon as the driver corrects the fault, the Advisory event shall disappear automatically from the screen.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.331	Advisory event shall be displayed on the IDU as long as the incorrect handling is active. It shall disappear automatically when the incorrect action is corrected.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_10.01.Zefiro-Europe.TRS.73
Description	It shall be possible for the driver, crew and maintenance personnel to see all active events on every IDU in the train (relevant for user category).
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.330	An active events list shall be provided for each user category, ie. driver or maintainer.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_10.01.Zefiro-Europe.TRS.130
Description	Alarm events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU before it is removed from the screen.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.329	Alarm events shall always require acknowledgement on the IDU, the alarm message shall be removed from the screen only after acknowledgement.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_10.01.Zefiro-Europe.TRS.110
Description	Warning events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU before it is removed from the screen.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.327	Warning events shall be presented on the IDU and they shall be acknowledged by pressing a button on the IDU before it is removed from the screen.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

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ID	2F_10.01.Zefiro-Europe.TRS.107
Description	Maintenance personal shall be able to acknowledge all events both individually and as a group except for serious fault events that shall be acknowledged individually.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.328	If the IDU is activated by a maintainer (maintanance mode), the personal shall be able to acknowledge all events both individually and as a group except for Serious fault events that shall be acknowledged individually.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.348	The driver shall be able to acknowledge all events once, when CCUO allowed to do that.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.4 Buzzer interface

3.1.4.1 Function Design

Critical events will sound a buzzer in the cab to notify the driver.

3.1.4.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.76
Description	New Alarm event shall trigger the buzzer in the active cab.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.228	Each new Alarm event shall give a trigger to the buzzer in the active cab. Triggers shall not be queued.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_10.01.Zefiro-Europe.CONCEPT.334	The Buzzer in the cab shall sound for 5s and for 1s every 5s for maximm 15s or until it is acknowledged by the user by selecting the acknowledge button on the event field on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

3.1.5 Function Condition data handling

3.1.5.1 Function Design

Condition data is used to store none volatile parameters and counters as defined in the different system SFRSes. The collection in this section only defines the base functionality to use for such condition data.

3.1.5.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.78
Description	Condition data of type parameters and counters shall be supported.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.94
Description	The diagnostic system shall also store other data (called condition data) about the various systems such as operation time and number of operations.
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.142	The TCMS system shall implement condition data (parameters and counters) as defined in the different systems connected to TCMS and defined in the system SFRSes (except for events from the propulsion system).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.304	Condition data of type parameters shall have a default value to be used as start value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.305	Condition data of type counters shall have a counter limit to be used to compare the counter value.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

3.1.6 Diagnostics data set at train level

3.1.6.1 Function Design

3.1.6.2 Requirements

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.345	Before every TCMS shutdown an event shall be set, in order to indicate that the train has been requested to power off	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_10.01.Zefiro-Europe.CONCEPT.346	When a user log in as a Maintainer in the IDU an event shall be set, in order to indicate that the train is under maintenance	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_10.01.Zefiro-Europe.CONCEPT.347	When a user log out from Maintainer in the IDU an event shall be set, in order to indicate that the train is no more under maintenance	Derived Requirement	4S_09.03.05.-.TCMS Software		

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3.2 Function Software version handling

3.2.1 Function Design

The version reporting system consists of a server that stores version information reported by the different TCMS devices. The TCMS application software, by use of information provided by connected systems, reports also version information for those. The IDU and MCG additionally include requester to get version information from the server. The propulsion system reports their version directly to the server.

3.2.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.80
Description	The TCMS system shall include a version reporting system used for application software version (for presentation on the IDU).
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.244	The TCMS system shall include a VRS server.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.245	The TCMS controllers shall include VRS client and software/hardware reporters to send version/revision data to the VRS server for the individual controller.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.291	The TCMS system shall implement generic VRS reporters for MIO-units.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.306	The TCMS system shall implement generic VRS reporters for systems connected to TCMS, to report software versions, serial number and hardware revisions. **Note:** Information defined in the different system Interface Control Documents.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.307	The propulsion system components (DCU2/x, LIM and MIO) shall report their software/hardware information directly to the TCMS VRS server over IPT. Note: For DCU2s this shall be done by instantiating the VRS client and software/hardware reporters to each DCU2. For LIM and MIO the general VRS reporters needs to be implemented in the application software.	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.308	The IDU shall include a VRS requester to access the stored VRS data.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.117
Description	The stored software versions shall be possible to upload to a connected service LapTop and Remote (via MCG).
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.246	It shall be possible to upload the stored VRS data from a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.247	The MCG shall provide a service to upload VRS data from the train.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

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3.3 Maintenance

3.3.1 Service Access

3.3.1.1 Function Design

Service access is performed from a service laptop PC connected to a service access point on an IP ring switch.

3.3.1.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.116
Description	From Each cab IP service port there is possible to upload diagnostic data, download software, check communication, configuration, software version numbers, monitor and temporary changing parameters and initiate function tests.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.259	In each cab there shall be an easy accessible service port connected to an IP switch.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_10.01.Zefiro-Europe.CONCEPT.332	From Each IP service port it shall be possible to upload diagnostic data, download software, check communication, configuration, software version numbers, monitor and temporary changing parameters and initiate function tests.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.119
Description	It shall be possible to view signals and parameters from a connected service LapTop.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.120
Description	It shall be possible to temporarly change parameters in the TCMS software from a connected service LapTop.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.123
Description	It shall be possible to analyze signals status over time in graphical format.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.285	It shall be possible to view signals and parameters in all devices applications, using DCUTerm on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.316	It shall be possible to temporary change parameters in the CSS-based devices applications, using DCUTerm on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.317	DCUTerm shall support logmonitor, fast log and trace window to enable analysis of signals over time in graphical format.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.318	DCUTerm shall support GUI in the following languages Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

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3.3.2 Software Download

3.3.2.1 Function Design

Download of software to the TCMS system and to verify that correct files are downloaded.

3.3.2.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.61
Description	It shall be possible to download application software and other information to the TCMS units (including the DCU/A, /F, /L, /M and LIM) from a connected service LapTop.
Safety level	SIL 0

ID	2F_10.01.Zefiro-Europe.TRS.122
Description	The MTVD tool shall be used for downloading and version check of downloaded applications.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.269	It shall be possible to download software (operating system, applications, configuration files) to the Mitrac controllers connected to ECN, using MTVD on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.320	The propulsion system shall include needed functionality to support software download over ECN to all included devices. Note: This also includes the LIM	Derived Requirement	4S_05.-.Propulsion		N/A
2F_10.01.Zefiro-Europe.CONCEPT.309	It shall be possible verify that correct versions have been dowloaded to the controllers.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.310	It shall be possible to check what versions are downloaded to the controllers without performing a new download operation.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.311	MTVD shall support GUI in the following languages Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3.3 Function Upload and analysis

3.3.3.1 Function Design

Upload of diagnostic data and performing analysis of the uploaded data.

3.3.3.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.95
Description	It shall be possible to perform remote upload of the stored diagnostic data to an off-board system.
Safety level	N/A

ID	2F_10.01.Zefiro-Europe.TRS.46
Description	The event information stored in the train diagnostic database shall be possible to upload to an off-board maintenance and service system for long term storage and additional off line analysis.
Safety level	N/A

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ID	2F_10.01.Zefiro-Europe.TRS.112
Description	The diagnostic system shall interface the Train Telediagnostica System for remote upload of stored event data.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.271	Upload of the TDS data using the TDS-Tool Protocol shall be supported.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.312	The TDS upload tool shall support GUI in the following languages: Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.333	Upload of diagnositc data to the Telediagnostica application shall be supported.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.121
Description	The MAVIS tool shall be used for analysis of uploaded events on a sevice LapTop.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.272	It shall be possible to upload TDS data, using TDS uploader on a service LapTop connected to ECN via a service access point.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.313	It shall be possible to analyse the uploaded data on the service LapTop using MAVIS.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A
2F_10.01.Zefiro-Europe.CONCEPT.314	MAVIS shall support GUI in the following languages: Italian and English.	Derived Requirement	4S_09.03.06.-.TCMS Hardware		N/A

3.3.4 Function Subscription of signals

3.3.4.1 Function Design

Base functions for the IDU as well as train to wayside communication for accessing signals from the TCMS system.

3.3.4.2 Requirements

ID	2F_10.01.Zefiro-Europe.TRS.115
Description	It shall be possible to subscribe to sub-system signals on MVB and IPT, e.g. from the IDU.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.273	IDU shall use IPT communication to subscribe to sub-system signals.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_10.01.Zefiro-Europe.CONCEPT.315	Signals sent to IDU from MVB shall be transferred on IPT by the CCU-O application.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

ID	2F_10.01.Zefiro-Europe.TRS.114
Description	A TSSP server shall be included to provide the functionality to subscribe to onboard signals from a ground based system (via the Train Telediagnostica System).
Safety level	N/A

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Nr	Description	Type	Derived to	User Interface	SIL
2F_10.01.Zefiro-Europe.CONCEPT.274	It shall be possible to subscribe to onboard MVB and IPT signals by the use of a TSSP server from a ground based system (via the Train Telediagnostica System).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

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4 Signal Interface

4.1 I/O Interface

4.1.1 Supplier system

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_10.01.Zefiro-Europe.CONCEPT.28			

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_10.01.Zefiro-Europe.CONCEPT.29			

4.1.2 TCMS SW

Inputs:

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_10.01.Zefiro-Europe.CONCEPT.287	Buzzer	L	2F_10.01.Zefiro-Europe.CONCEPT.228

4.2 Bus Interface

Supplier to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_10.01.Zefiro-Europe.CONCEPT.32			

TCMS SW to Supplier:

Nr	Functional name	Type	Linked to requirement
2F_10.01.Zefiro-Europe.CONCEPT.33			

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5 References

3EVT12158-VZI TDS Process

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6 Revision Log

6.1 Revision log 3EST000225-7918_B

Revision:	**_B**		
Date:	**2018-05-16**		
Editor:	**G. Parisi**		
Change ID DOORS	**Refers to requirement ID**	**Title**	**Change content**
2F_10.01.ISSUES.-.Zefiro-Europe.11	[2F_10.01.Zefiro-Europe.CONCEPT.293]: [2F_10.01.Zefiro-Europe.CONCEPT.294]:	Process ID PPC	PPC needs process ID from 180-255. Range increased from 200-255
2F_10.01.ISSUES.-.Zefiro-Europe.14	[2F_10.01.Zefiro-Europe.CONCEPT.345]: [2F_10.01.Zefiro-Europe.CONCEPT.346]: [2F_10.01.Zefiro-Europe.CONCEPT.347]:	Diagnostic events	New diagnostic events: • Before every TCMS shutdown an event shall be set, in order to indicate that the train has been requested to power off • When a user log in as a Maintainer in the IDU an event shall be set, in order to indicate that the train is under maintenance • When a user log out from Maintainer in the IDU an event shall be set, in order to indicate that the train is no more under maintenance
2F_10.01.ISSUES.-.Zefiro-Europe.15	[2F_10.01.Zefiro-Europe.CONCEPT.348]:		The acknowledge all events shall possible for the driver.

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